

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech Branch : Civil Engineering /Civil and Infrastructure Engineering

Semester : IV

Subject Code & Name: BTCVC402/BTCIC402 Environmental Engineering

Max Marks: 60

Date: 21/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- | | | |
|---|--|---|
| 1 | Which one of the following methods gives the best estimate of population growth of a community with limited land area for future expansion? | 1 |
| | a. Arithmetical increase method b. Geometrical increase method c. Incremental increase method d. Logistic method | |
| 2 | If the average daily water consumption of a city is 24000 cum, the peak hourly demand (of the maximum day) will be | 1 |
| | a. 1000 cum/h b. 1500 cum/hr c. 1800 cum/hr d. 2700 cum/hr | |
| 3 | The threshold odour number (TON) for a water sample of 40 ml, diluted to standard 200 ml mixture, in which odour is just barely detectable to the sense of smell, is | 1 |
| | a. 8 b. 5 c. 50 d. None of these | |
| 4 | Which of the following chemical compounds can be used for dechlorination of water? | 1 |
| | a. carbon dioxide b. bleaching powder c. sulphur dioxide d. chloramines | |
| 5 | The maximum permissible limit For fluoride in drinking water is | 1 |
| | a. 0.1 mg/l b. 1.5 mg/l c. 5 mg/l d. 10 mg/l | |
| 6 | The valve, which allows the flow only in one direction, is a | 1 |
| | a. reflux valve b. sluice valve c. gate valve d. None of these | |

- 7 Double filtration is used 1
- a. to increase the filtration capacity of slow sand filters b. to increase the filtration capacity of rapid sand filters c. for isolated colonies of a few houses d. for none of the above
- 8 The formula, which is most appropriate for the design of pressure pipes is 1
- a. Darcy Weisbach formula b. Mannings formula c. Chezy's formula d. Dupuit's formula
- 9 Which one of the following solid waste disposal methods is ecologically most acceptable? 1
- a. Sanitary landfill b. Incineration c. Composting d. Pyrolysis
- 10 Allowable disposable rate of application of sludge on land is determined by 1
- a. carbon content of sludge b. nitrogen content of sludge c. phosphorus content of sludge d. potassium content of sludge
- 11 Symbiosis, the beneficial association between algae and bacteria is used for treatment of waste water in the following unit? 1
- a. Activated Sludge b. Rotating Biological Disc c. Anaerobic Digester d. Oxidation Pond
- 12 The relative stability of a sewage sample, whose D.O. equals the total oxygen required to satisfy its BOD, is 1
- a. zero b. 1% c. 100% d. infinity

Q. 2 Solve the following.

12

- A) Explain various chemical characteristics of water? What are the various methods of determining the chemical characteristics? BT2/CO2 6
- B) What are various processes required to remove the various types of impurities? Give a neat sketch showing the various processes required in treating a river water, starting from the source of water. BT2/CO3 6

Q.3 Solve the following.

12

- A) What are the requirements of good distribution systems? Describe in brief various types of distribution systems. BT1/CO2 6
- B) What are the various chemical coagulants which are commonly used in coagulation process? How they remove suspended impurities? BT2/CO2 6

Q. 4 Solve Any Two of the following.

12

A) Write a short note on

BT3/CO2

6

a. Break point chlorination

b. Free available chlorine

c. Chlorine demand

B) Explain various physical and Chemical characteristics of wastewater.

BT2/CO2

6

C) Explain clearly how does a rapid gravity filter differ in its action from a slow sand filter?

BT3/CO3

6

Q.5 Solve Any Two of the following.

12

A) Explain Primary & Secondary treatment of waste water in detail.

BT2/CO1

6

B) Differentiate between BOD and COD.

BT2/CO3

6

C) In a BOD test, 1.0 ml of raw sewage was diluted to 100 ml and the dissolved oxygen concentration of diluted sample at the beginning was 6 ppm and it was 4 ppm at the end of 5-day incubation at 20°C. Find out the BOD of raw sewage.

BT4/CO3

6

Q. 6 Solve Any Two of the following.

12

A) What are the harmful effects of polluted air on human beings?

BT2/CO4

6

B) Define solid waste. Why it is necessary to manage solid waste. Explain various classification of solid waste based on source generation and characteristics.

BT2/CO4

6

C) Explain the following atmospheric condition:

BT2/CO4

6

a. Super adiabatic and Sub adiabatic

b. Neutral & Inversion

How do they influence the dispersion of pollutants in the atmosphere?

***** End *****